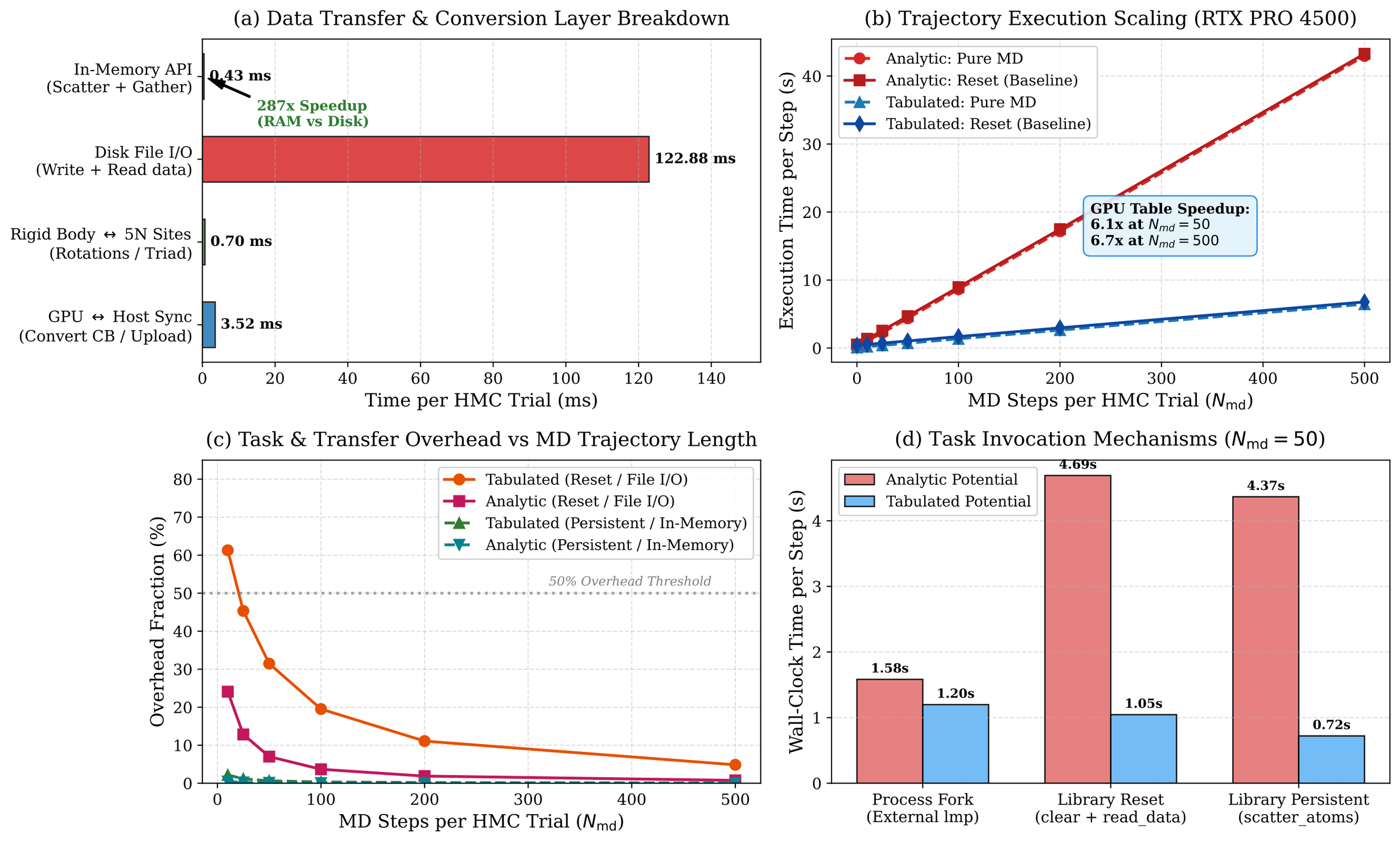

(a) Data Transfer & Conversion Layer Breakdown
In-Memory API
(Scatter + Gather)
0.43 ms
287x Speedup
(RAM vs Disk)
Disk File I/O
(Write + Read data)
122.88 ms
Rigid Body ↔ 5N Sites
(Rotations / Triad)
0.70 ms
GPU ↔ Host Sync
(Convert CB / Upload)
3.52 ms
Time per HMC Trial (ms)
(b) Trajectory Execution Scaling (RTX PRO 4500)
Analytic: Pure MD
Analytic: Reset (Baseline)
Tabulated: Pure MD
Tabulated: Reset (Baseline)
GPU Table Speedup:
6.1x at $N_{md} = 50$
6.7x at $N_{md} = 500$
Execution Time per Step (s)
MD Steps per HMC Trial (N_{md})
(c) Task & Transfer Overhead vs MD Trajectory Length
Tabulated (Reset / File I/O)
Analytic (Reset / File I/O)
Tabulated (Persistent / In-Memory)
Analytic (Persistent / In-Memory)
50% Overhead Threshold
Overhead Fraction (%)
MD Steps per HMC Trial (N_{md})
(d) Task Invocation Mechanisms ($N_{md} = 50$)
Analytic Potential
Tabulated Potential
1.58s
1.20s
4.69s
1.05s
4.37s
0.72s
Wall-Clock Time per Step (s)
Process Fork
(External lmp)
Library Reset
(clear + read_data)
Library Persistent
(scatter_atoms)